

22317

25226

3 Hours / 70 Marks

Seat No.

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- Instructions :**
- (1) All Questions are *compulsory*.
 - (2) Answer each next main Question on a new page.
 - (3) Illustrate your answers with neat sketches wherever necessary.
 - (4) Figures to the right indicate full marks.
 - (5) Assume suitable data, if necessary.
 - (6) Mobile Phone, Pager and any other Electronic Communication devices are not permissible in Examination Hall.

Marks

1. Attempt any FIVE of the following :

10

- (a) List any four operations on data structure.
- (b) Define Recursion.
- (c) Define the terms related to Binary tree.
 - (i) Leaf node
 - (ii) Level of node
- (d) Enlist any four operations of queue.
- (e) Define undirected graph with diagram.
- (f) Define space complexity with example.
- (g) Differentiate between Stack and Queue. (any four points)



2. Attempt any THREE of the following :**12**

- (a) Find the position of element 88 using binary search method in an array given below :

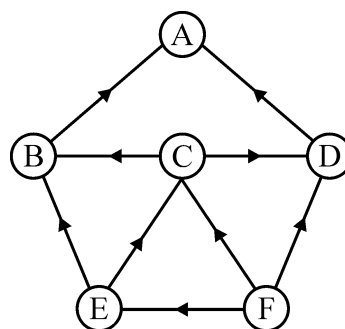
$$A = \{77, 33, 44, 11, 88, 22, 66, 55\}$$

- (b) Describe the concept of linked list with terminologies : Node, Next Pointer, NULL Pointer and Empty list.
- (c) Write a procedure to insert an element into a queue and delete an element from a queue.
- (d) Draw a tree structure of the following expression :

$$(a - 2b - 5c)^2 (4d - 6e)^5$$

3. Attempt any THREE of the following :**12**

- (a) Differentiate between linear data structure and non-linear data structure. (Min. 4 points)
- (b) Evaluate the following postfix expression :
A : 10, 2, *, 15, 3, /, +, 12, 3, +, +. Show diagrammatically each step of evaluation using stack.
- (c) Sort the following numbers in ascending order using Bubble sort. Given Number : {16, 23, 13, 9, 7, 5} and write the output after each iteration.
- (d) For the following graph :
- Give adjacency matrix representation
 - Give adjacency list representation

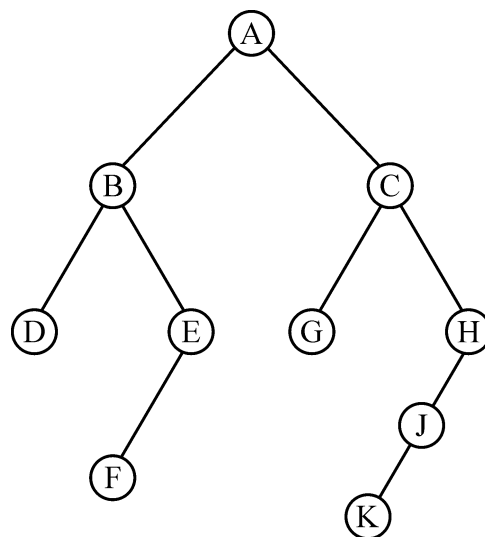
**Fig. 1**

4. Attempt any THREE of the following :**12**

- (a) Describe the working of radix sort with example.
- (b) Draw a binary search tree using following elements :
35, 15, 40, 7, 100, 28, 82, 53, 25, 3
- (c) Write a program to traverse a linked list.
- (d) Convert infix string $((A + B) * (C - D)) / (E + F)$ into prefix string with stack.
- (e) Write an algorithm to insert new node at the beginning of a singly linked list.

5. Attempt any TWO of the following :**12**

- (a) Show the effect of PUSH and POP operations on the stack of size 10. The stack contains 40, 30, 52, 86, 39, 45, 50 with 50 being at the top of the stack. Show diagrammatically the effect of
 - (i) PUSH 59
 - (ii) PUSH 85
 - (iii) POP
 - (iv) POP
 - (v) PUSH 59
 - (vi) POP
- (b) What is binary tree ? For given tree perform preorder, postorder and inorder traversal.

**Fig. 2**

- (c) Create a singly linked list using data fields 90, 25, 46, 39, 56 search a node 40 from the singly linked list (SLL) and show procedure step-by-step with the help of diagram from start to end.

6. Attempt any TWO of the following :

12

- (a) Describe the working of selection sort method. Also sort given input list in ascending order using selection sort.

Input list : 16, 22, 12, 56, 65, 78

- (b) Convert the following Infix expression to its postfix expression using stack and show the details of stack at each step of conversion

Expression :

$a \uparrow b * c - d + e / f / (g + h)$

Note –

$\uparrow$ indicates exponent operator

- (c) Differentiate between linear linked list and circular linked list. (Min. 6 points)
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